Stock Market Value Prediction System using RNN

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***Abstract*—Predicting stock prices is a long-standing and challenging problem in financial analytics due to the highly dynamic, nonlinear, and stochastic nature of financial markets. Accurate forecasting models can support informed investment decisions and risk management; however, market volatility and external influences often limit predictive reliability. In this project, we investigate a range of machine learning and deep learning approaches for stock price prediction using historical time-series data. The dataset consists of normalized stock market features, including open, high, low, close, adjusted close, and trading volume. Multiple architectures are implemented and compared, including Simple Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), Gated Recurrent Units (GRU), Transformer Encoders, and a hybrid CNN-BiLSTM with an attention mechanism, along with baseline models such as a Random Walk predictor and an over-regularized deep LSTM. Model performance is evaluated using several regression and trend-based metrics, including RMSE, MAE, R^2 score, maximum error, and directional accuracy to assess both magnitude and trend prediction quality. The results indicate that while deep learning models can effectively capture temporal patterns in stock price movements, increased model complexity does not always lead to improved performance in volatile market conditions. This work is presented as a work-in-progress, with potential improvements and future research directions discussed.**

***Index Terms*—Stock price prediction, time series forecasting, recurrent neural networks, long short-term memory, gated recurrent unit, deep learning, financial data analysis.**

I. Introduction

Stock markets play a central role in modern economies, influencing investment strategies, corporate valuation, and economic policy. The ability to predict stock price movements has significant practical importance for investors, portfolio managers, and financial institutions. However, stock price prediction remains a difficult task due to the inherent uncertainty, noise, and nonlinearity present in financial time-series data.

Traditional financial theories, such as the Efficient Market Hypothesis (EMH), suggest that stock prices fully reflect all available information, implying that consistent prediction of future prices is extremely difficult. Despite this, empirical evidence shows that historical price patterns may still contain useful information, particularly when analyzed using advanced computational models.

Recent advances in machine learning and deep learning have enabled the development of predictive models capable of learning complex temporal patterns from time-series data. These models do not rely on strict statistical assumptions and can adapt to nonlinear relationships within the data. This project investigates the effectiveness of such models in predicting stock prices based on historical market data. The project is particularly interesting as it combines real-world financial data with practical applications of time-series modeling, data preprocessing, and performance evaluation.

The primary goal of this study is to analyze and compare the performance of multiple sequence-based learning architectures for stock price prediction. These include recurrent neural network variants such as Simple RNN, LSTM, and GRU, as well as more advanced models such as Transformer Encoders and hybrid CNN-BiLSTM architectures with attention mechanisms. In addition, baseline models are included to provide performance context. By evaluating prediction accuracy using multiple regression and directional metrics, this project aims to highlight the strengths and limitations of each approach and demonstrate that increased model complexity does not necessarily guarantee improved performance in financial time-series forecasting.

II. Related Work and Literature Review

Stock price prediction has been widely studied across multiple disciplines, including economics, statistics, and computer science. Early approaches primarily relied on statistical and econometric models such as Linear Regression, Autoregressive Integrated Moving Average (ARIMA), and GARCH models. While these techniques offer interpretability and theoretical grounding, they often assume linearity and stationarity, which rarely hold in real-world financial markets [1].

With the emergence of machine learning, researchers began exploring non-linear models such as Support Vector Machines (SVMs), Decision Trees, and Random Forests for stock price prediction. These models demonstrated improved performance over traditional statistical approaches by capturing nonlinear dependencies in market data [2]. However, many of these methods still require extensive feature engineering and struggle to model long-term temporal dependencies inherent in financial time series.

More recently, deep learning techniques have gained prominence in financial time-series forecasting. Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks are particularly well-suited for sequential data and have

been shown to outperform classical methods in many stock prediction tasks [3]. LSTM networks address the vanishing gradient problem and can capture long-term dependencies in time-series data, making them a popular choice for financial forecasting. Variants such as Gated Recurrent Units (GRU) further reduce model complexity while maintaining comparable performance. Gated Recurrent Units (GRUs) simplify the LSTM architecture by combining the forget and input gates, allowing efficient learning with fewer parameters [5].

In addition to recurrent architectures, attention-based and hybrid models have been explored to improve temporal representation learning. Convolutional recurrent hybrids and attention mechanisms have been proposed to capture both local price patterns and long-range dependencies. More recently, Transformer-based architectures, which rely entirely on self-attention mechanisms, have been applied to financial time series with mixed results, often requiring large datasets and careful tuning to outperform recurrent models.

Despite promising results, several challenges remain. Deep learning models often require large datasets, are computationally expensive, and lack interpretability. Additionally, many studies highlight that model performance can degrade significantly during periods of high market volatility or unexpected economic events [4]. These limitations motivate continued research into model comparison, architectural suitability, hybrid approaches, and evaluation metrics that better reflect practical forecasting performance.

III. Dataset and Methodology

A. Dataset Description

The dataset used in this project consists of historical stock price data collected over a specified time period. Each data record contains the following features:

- Opening price
- Closing price
- Highest price of the day
- Lowest price of the day
- Trading volume

These features provide a comprehensive representation of daily stock market activity and are commonly used in financial time-series prediction tasks. The inclusion of both price-based and volume-based features allows the models to capture market trends, volatility, and trading behavior.

B. Data Preprocessing

Raw financial data often contains noise, missing values, and varying scales across features. To address these issues, the following preprocessing steps were applied:

- Removal or handling of missing values
- Normalization or scaling of numerical features
- Conversion of raw data into time-series sequences using sliding windows

The sliding window transformation converts historical price data into supervised learning samples, where a sequence of past observations is used to predict the next time step. This representation enables the models to learn temporal dependencies and sequential patterns inherent in stock price movements.

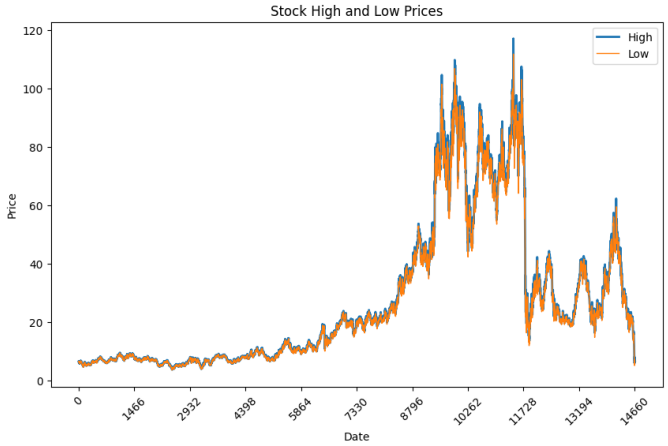


Fig. 1. 'High' and 'Low' prices from the stock data.

C. Model Training

The dataset was split into training and testing sets to evaluate each model's generalization ability. All models were trained on historical price sequences and optimized using the Mean Squared Error (MSE) loss function. Training was performed for a fixed number of epochs to ensure fair comparison across architectures, and the temporal order of the data was preserved during training. Model performance was evaluated on unseen test data using multiple regression-based and directional metrics.

IV. Experimental Results

The trained models were evaluated on unseen test data to assess their predictive performance and generalization ability. Multiple quantitative evaluation metrics were used, including Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE), R^2 score, maximum error, and directional accuracy. These metrics collectively measure prediction accuracy, robustness to large errors, explanatory power, and the ability to correctly predict the direction of price movement.

TABLE I
Performance Comparison of Prediction Models on the Test Dataset

Model	RMSE	MAE	MAPE (%)	R^2	Max Err.	Dir. Acc.
Simple RNN	0.0220	0.0161	22.39	0.9970	0.1201	0.7303
LSTM	0.0496	0.0452	71.81	0.9846	0.1407	0.7774
GRU	0.0212	0.0152	19.20	0.9972	0.1614	0.6890
Transformer Encoder	0.0702	0.0497	74.02	0.9692	0.5274	0.5425
CNN-BiLSTM-Attention	0.0299	0.0209	30.36	0.9944	0.2116	0.6439
Random Walk	1.7951	1.7545	2434.26	-19.1891	2.6567	0.5063
Over-Regularized Deep LSTM	0.2644	0.2428	314.20	0.5619	0.9620	0.4978

Overall, the experimental results show that learning-based models are able to capture general stock price trends and perform well during relatively stable market periods. Recurrent architectures such as LSTM and GRU achieve lower prediction

error compared to simpler models, while more complex architectures such as Transformer Encoders and CNN-BiLSTM with attention demonstrate mixed performance depending on market conditions. The attention mechanism enables the model to focus on the most relevant time steps when making predictions, improving representation of complex temporal dependencies [6]. Baseline models, including the Random Walk and the over-regularized deep LSTM, perform significantly worse, providing useful lower-bound references for comparison. We decided to use Random Walk baseline as it aligns with the Efficient Market Hypothesis, which suggests that price movements are largely unpredictable and resemble stochastic processes [7].

Visual comparisons between actual and predicted prices further support these findings. As shown in the full test-set plots, most learning-based models closely follow the overall direction of the true price series. However, during periods of high volatility, prediction accuracy decreases, and deviations between predicted and actual values become more pronounced. This behavior highlights the inherent difficulty of forecasting abrupt market changes using historical price data alone.

To better examine short-term dynamics, zoomed-in plots of selected test intervals were also analyzed. These visualizations reveal small lag effects and smoothing behavior across most models, particularly during rapid price fluctuations. While such smoothing helps reduce noise, it can limit responsiveness to sudden market movements. Directional accuracy analysis indicates that several models are able to predict the general direction of price changes more reliably than random baselines, even when magnitude errors increase.

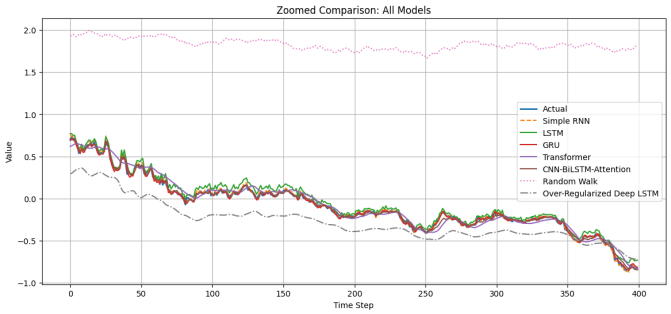


Fig. 2. Comparison of all model predictions.

Visual analysis indicates that while most learning-based models closely follow the overall direction of true stock prices, small lag effects and smoothing behavior are consistently observed, particularly during periods of rapid price change. This behavior is characteristic of time-series forecasting models trained on historical data, as they rely on past observations to infer future values and therefore respond more conservatively to sudden market shifts. Models with stronger regularization or deeper architectures tend to exhibit increased smoothing, which helps reduce noise but can limit responsiveness to short-term volatility. Conversely, simpler recurrent models often react more quickly but may produce fewer stable predictions. These observations highlight an important trade-off between trend-following stability and short-term sensitivity in stock price prediction, reinforcing the challenge of accurately modeling highly volatile financial markets using historical data alone.

V. Discussion

The experimental results demonstrate that machine learning-based models can effectively learn meaningful patterns from historical stock price data. However, the findings also highlight the inherent limitations of purely data-driven approaches in financial forecasting.

One key observation is reduced performance during periods of abrupt market changes. This limitation arises because historical price data alone cannot fully capture external influences such as economic announcements, geopolitical events, or investor sentiment. Furthermore, deep learning models tend to smooth predictions, which may reduce responsiveness to sudden price movements. This smoothing effect is visible in the full and zoomed plots and is reflected by increases in error metrics during volatile intervals, even when overall trend tracking remains strong.

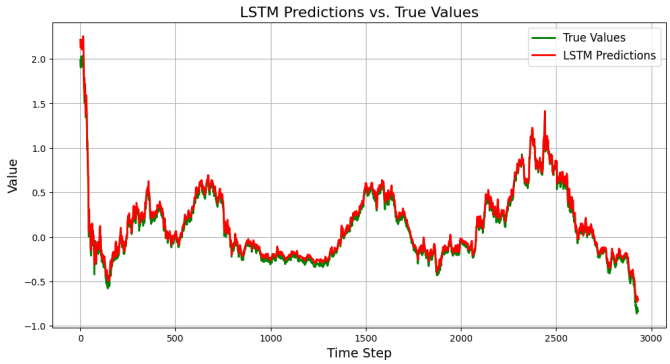


Fig. 3. Best Model Performance.

Across the seven models evaluated, recurrent architectures (LSTM and GRU) generally provide the best balance between stability and accuracy. Their gating mechanisms help retain relevant information across time steps, which improves performance relative to a Simple RNN that lacks explicit memory control. In contrast, the Transformer Encoder, despite its architectural complexity, did not consistently outperform recurrent models in this setting. This outcome aligns with the fact that attention-based models often require larger datasets and careful tuning to realize their full advantage, while short input windows and noisy financial series can limit the benefit of purely attention-driven representations.

The CNN-BiLSTM with Attention model introduces additional representational power by combining convolutional feature extraction, bidirectional recurrence, and attention-based weighting across time steps. While this hybrid architecture can capture both local fluctuations and broader temporal structure, the results indicate that added complexity does not necessarily guarantee improved performance. In particular, models with

higher capacity may introduce additional smoothing or overfitting risk without a corresponding gain in generalization, emphasizing the importance of model suitability rather than model size alone.

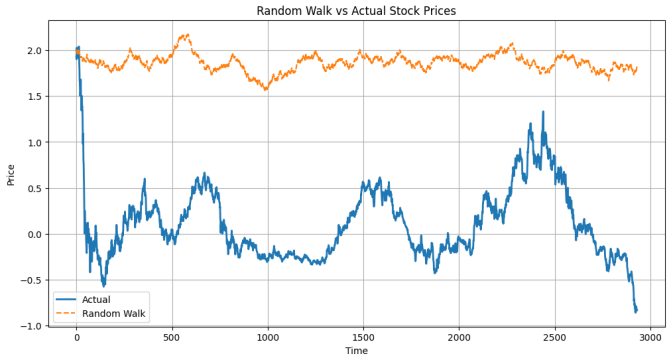


Fig. 4. Worst Model Performance.

The inclusion of baseline and intentionally weak models strengthens the evaluation by providing reference points for interpretation. The Random Walk baseline illustrates the difficulty of the task and provides a lower-bound comparison under a stochastic assumption, while the over-regularized deep LSTM demonstrates how excessive regularization can prevent a model from learning meaningful temporal dynamics, even when the architecture itself is complex. These baselines support the conclusion that strong performance depends not only on architectural sophistication but also on balanced training design and appropriate inductive bias for time-series data.

Finally, the metrics used provide complementary perspectives on performance. RMSE and MAE quantify magnitude-based prediction error, maximum error highlights worst-case deviations that may be critical in financial settings, and directional accuracy evaluates whether models correctly predict the direction of movement, which is often relevant for decision-making. Taken together, these results suggest that while deep learning models can capture useful temporal structure in stock price sequences, forecasting performance remains constrained by market volatility and unobserved external drivers. Despite these challenges, the results are encouraging for a work-in-progress study, and the comparative analysis establishes a strong foundation for future improvements such as incorporating additional market indicators or external information sources.

VI. Future Work

Several directions can be explored to improve the performance and robustness of the proposed approach. First, additional feature engineering could be incorporated beyond raw OHLCV inputs. This includes technical indicators such as RSI, MACD, Bollinger Bands, and volatility measures, which may provide more informative signals than prices alone. While simple moving averages were explored as baselines, richer indicators could be integrated as model inputs rather than used only for comparison.

Second, the framework could be extended by incorporating external information sources that influence market movements but are not present in historical price data. Examples include sentiment signals extracted from financial news, earnings reports, macroeconomic announcements, and social media, as well as broader market indices (e.g., S&P 500) or sector-level indicators. Including these exogenous variables may improve performance particularly during high-volatility periods where price history alone is insufficient.

Third, additional modeling improvements could be explored through systematic hyperparameter tuning (e.g., sequence length, hidden units, dropout, learning rate, and batch size) and stronger validation strategies such as walk-forward validation. This would provide a more reliable estimate of generalization performance for time-series data than a single split. Further architectural extensions could also be examined, such as deeper or multi-head attention variants, bidirectional recurrent models (with careful discussion of forecasting realism), and ensemble approaches that combine outputs from multiple architectures.

Finally, the current study focuses on single-step forecasting, which is often easier due to short-horizon autocorrelation. A natural extension is multi-step forecasting, where models predict multiple future days either iteratively or using sequence-to-sequence outputs. Evaluating performance across longer horizons would provide a more realistic assessment of model usefulness for decision-making and would further highlight the limitations of historical-price-only prediction in highly volatile financial markets.

VII. Conclusion

This project explored stock price prediction using machine learning techniques applied to historical financial time-series data. The results demonstrate that predictive models can learn temporal patterns and approximate future price movements, particularly during relatively stable market periods. At the same time, performance degrades during high-volatility intervals, reinforcing the difficulty of forecasting abrupt price changes from historical OHLCV features alone and highlighting the impact of unobserved external factors such as news, macroeconomic events, and market sentiment.

A central contribution of this work is a comparative evaluation of seven models with varying complexity, including Simple RNN, LSTM, GRU, Transformer Encoder, and a hybrid CNN-BiLSTM with attention, along with two baseline models (Random Walk and an over-regularized deep LSTM) that provide meaningful performance context. Across the evaluated metrics, RMSE, MAE, MAPE, R^2 score, maximum error, and directional accuracy, gated recurrent architectures such as LSTM and GRU generally offered the most reliable balance of predictive accuracy and stability. In contrast, more complex architectures did not consistently produce superior results, and intentionally weak baselines confirmed that both stochastic assumptions (Random Walk) and excessive regularization can significantly degrade forecasting quality. These findings emphasize that model suitability and training design are often

more important than architectural complexity when working with noisy and non-stationary financial data.

As a work-in-progress, this study highlights both the potential and limitations of machine learning-based stock prediction and provides a strong foundation for future improvement. Extensions such as incorporating richer technical indicators, adding exogenous variables (e.g., sentiment and macroeconomic signals), applying more rigorous time-series validation, and expanding to multi-step forecasting may further improve robustness and provide deeper insight into real-world market behavior.

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